

Forecasting Remaining Useful Life (RUL): A Data Science Solution for Proactive Aircraft Engine Maintenance

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1.1 Introduction

Predictive Maintenance (PdM) is recognized as essential for raising aircraft safety, cutting downtime, cutting cost, and optimising maintenance schedules [10]. Remaining Useful Life (RUL) is fundamentally defined as the time or usage cycles a system or component has left before it ceases to operate while maintaining the necessary performance and safety standards . Accurate RUL forecasting is vital as it aids in preventing expensive unplanned repairs, increasing component efficiency, maintaining safety, increasing aircraft availability, and reducing costs [11].

Prognostics and Health Management (PHM) are considered crucial in the industrial big data era because they improve the accuracy of future failure predictions, thus reducing expenses associated with inventory, maintenance, and labour [4]. Traditional maintenance approaches, based on fixed schedules or simple usage thresholds, are often inadequate, potentially leading to unscheduled downtime and inefficient resource allocation [19].

The implementation of Internet of Things (IoT) technology is an essential element of PdM in manufacturing, as machines equipped with sensors can collect data, allowing predictive models to detect errors/failures that might otherwise be missed by human observation [24]. Several studies in prognostics rely on the NASA Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) dataset, an open-source benchmark featuring simulated turbofan engine units under realistic flight circumstances [4]. This dataset incorporates damage propagation modelling that takes into consideration actual flying circumstances and connects the degradation process to its operational history [4].

1.2 Problem Definition

Unexpected machine failures in industrial environments, especially in aviation, lead to high maintenance costs, unplanned downtime, and safety risks [1]. Turbofan engines are critical components, and their faults account for approximately 60% of all faults encountered in modern aircraft systems [10]. Predicting RUL is highly beneficial, but remains the most critical task for predictive maintenance [10].

Traditional maintenance techniques, such as reactive or preventive (time-based) maintenance, frequently lack cost-effectiveness and fail to predict breakdowns, leading to unplanned outages [1]. The complexity of RUL prediction stems from several factors, including the high dimensionality and non-linearity of multi-source monitoring parameters, as well as the difficulty in accurately modeling the equipment degradation process [5]. Furthermore, deep learning methods used for RUL prediction often face challenges due to the scarcity of target domain samples and the lack of explicit dynamic equations describing the degradation process [30].

1.3 The recommended solution

The recommended solution is a data-driven approach leveraging advanced machine learning (ML) and deep learning (DL) models combined with effective data preparation and pre-processing steps. Suggested solutions include:

1. A model combining effective pre-processing steps with a Long Short-Term Memory (LSTM) network to improve RUL prediction accuracy on the C-MAPSS turbofan engine dataset.

2. A Deep Learning Model (CNN-LSTM-Attention) designed to enhance feature recognition and achieve precise predictions by sequentially employing Convolutional Neural Networks (CNN) for feature extraction and an LSTM network augmented with attention mechanisms.
3. A hybrid predictive model combining eXtreme Gradient Boosting (XGBoost) and Neural Networks (NN), particularly useful for machine failure forecasting by addressing class imbalance and enhancing problem detection.
4. A combined approach using Bidirectional Long Short-Term Memory (BiLSTM) networks integrated with feature-selection algorithms and change point detection mechanisms to handle varied operational conditions.
5. A Deep Reinforcement Learning (DRL) framework integrated with probabilistic RUL prognostics (estimated via CNNs with Monte Carlo dropout) to adaptively plan maintenance actions.

1.4 Project Scope

The scope focuses on enhancing the capability and accuracy of RUL prediction models for complex aviation machinery.

1.4.1 Aims

The principal aims of such projects include:

- Improving RUL prediction accuracy and performance for aircraft turbofan engines.
- Enhancing safety, efficiency, and reliability of turbofan engine maintenance.
- Minimizing the effort and cost associated with maintenance processes.
- Providing reliable scheduling options for predictive maintenance tasks.
- Developing models robust against complex issues like nonlinearity, high dimensionality, and noise.

1.4.2 Objectives

Key objectives associated with the proposed solutions are:

- Utilising the NASA C-MAPSS turbofan engine degradation dataset (subsets FD001, FD002, FD003, FD004) for model validation and benchmarking [2].
- Implementing effective pre-processing steps, such as correlation analysis to select relevant sensors, and scaling data using standard scaling or min-max normalization [29].
- RUL Labeling: Calculate and clip Remaining Useful Life using a piecewise linear function.
- Designing an improved degradation model, such as a piecewise linear degradation model or a clipped RUL function, to determine the starting point of deterioration and assign RUL target labels [2].
- Developing and training deep learning models (Deep LSTM, CNN-LSTM-Attention, BiLSTM) or hybrid ML models (XGBoost/NN, RFR, SVR) [1].
- Achieving minimum Root Mean Squared Error (RMSE) and Score function values, demonstrating improved performance compared to existing methods [2].
- Designing an LLM-driven conversational interface to allow users to interact with the RUL system through natural language and receive actionable insights.

1.5 Target user

The models developed for Remaining Useful Life (RUL) prediction of aircraft turbofan engines target several key groups involved in industrial operation [4], planning, and maintenance:

Primary Operational Users: The core target users are Aviation engineers and maintenance personnel. RUL prediction provides these engineers with the confidence needed to know how long the engine or its components can continue to operate effectively before reaching a critical state. This capability helps engineers schedule maintenance to avoid the risk of catastrophic failure, and plan and execute maintenance activities optimally [12].

Maintenance Management and Planning: Target users include personnel responsible for planning and decision-making, aiming to enhance organizational efficiency:

- Maintenance Planners and Schedulers utilize the prediction output for effective scheduling and to specify the optimal moment of engine replacement [19].
- Decision-makers use the results for adaptive and flexible maintenance planning and responding to urgent cases that require immediate maintenance [2].
- The application supports both the manufacturing and the servicing sectors of the aviation industry.

1.6 Methodology

The methodology for developing these predictive models is generally data-driven and follows sequential steps:

1. **Data Collection:** Collecting run-to-failure simulation data (C-MAPSS) or real-life historical data (pilot/maintenance reports, sensor readings) [11].
2. **Data Pre-processing:** Preparing the data through cleansing, standardizing, or imputing missing values. This includes tasks like data normalization (Min-Max or Standard Scaling) to bring values into a fixed range and RUL calculation [20].
3. **Feature Engineering/Selection:** Identifying and selecting the most effective parameters. This might involve manual selection, using algorithms like Correlation Analysis, or leveraging polynomial features. Some approaches construct a new feature based on the difference between sensor measurements [6] [9].
4. **Target Label Definition:** Calculating the RUL target feature, often using an improved piecewise linear degradation model to clip the RUL in the initial, healthy stages [2].
5. **Model Selection and Training:** Selecting and training appropriate regression models, such as Deep LSTM, Random Forest Regression (RFR), Support Vector Regression (SVR), or hybrid architectures like CNN-GRU or CNN-LSTM-Attention [3].
6. **Hyperparameter Tuning and Validation:** Optimising model parameters using techniques like Random Search CV or Bayesian optimization, often coupled with k-fold cross-validation (5-fold cross-validation) [22].
7. **Result Evaluation:** Assessing performance using metrics such as RMSE, R^2 score, MAE, and score function [2] [15].
- 8.

LLM-Based Maintenance Assistant (Planned Future Extension): the deployment of a natural-language conversational assistant that wraps the trained ensemble model to explain RUL predictions, generate maintenance recommendations, and support troubleshooting through plain-language interaction with engineers is identified as a priority future-work item (see Section 7.4).

1.7 Project Plan

1.7.1 Semester 1 – Phase 1: Data Preparation and Model Development

During the first semester, the focus will be on data understanding, preprocessing, and initial model development.

Key Activities:

- **Dataset Acquisition:** Obtain and preprocess datasets (e.g., NASA FD001 turbofan dataset from Kaggle).
- **Exploratory Data Analysis (EDA):** Visualize and analyze dataset characteristics to identify trends and anomalies.
- **Feature Engineering:** Apply normalization/scaling (StandardScaler, MinMax), feature selection (e.g., correlation analysis, ReliefF), and generate new features (e.g., polynomial or difference-based).
- **Model Selection and Training:** Begin developing baseline models using traditional machine learning algorithms such as Linear Regression (LR), Support Vector Regression (SVR), and Random Forest Regressor (RFR).
- **Initial Evaluation:** Perform model validation using techniques like K-Fold Cross Validation and metrics such as RMSE and R^2 .

Expected Outcome:

A well-prepared dataset and baseline predictive models that can accurately estimate RUL on validation data.

1.7.2 Semester 2 – Phase 2: Model Optimization and Evaluation

The second semester will focus on advanced model tuning, comparison, and result interpretation.

Key Activities:

- **Advanced Model Training:** Implement and train deep learning models such as LSTM, GRU, or CNN architectures for RUL prediction.
- **Hyperparameter Optimization:** Use RandomSearchCV, GridSearchCV, or Bayesian optimization for fine-tuning models.
- **Model Comparison:** Compare performance between machine learning and deep learning models.
- **Result Evaluation:** Calculate performance metrics (RMSE, R^2) and validate model reliability through cross-validation.
- **Interpretability and Analysis:** Use SHAP or LIME to explain model predictions and enhance interpretability.
- **Documentation and Presentation:** Prepare the final report, visualizations, and project presentation for submission.

Expected Outcome:

An optimized and interpretable predictive model with documented results and analysis, ready for deployment or publication.

1.8 The Gantt Chart

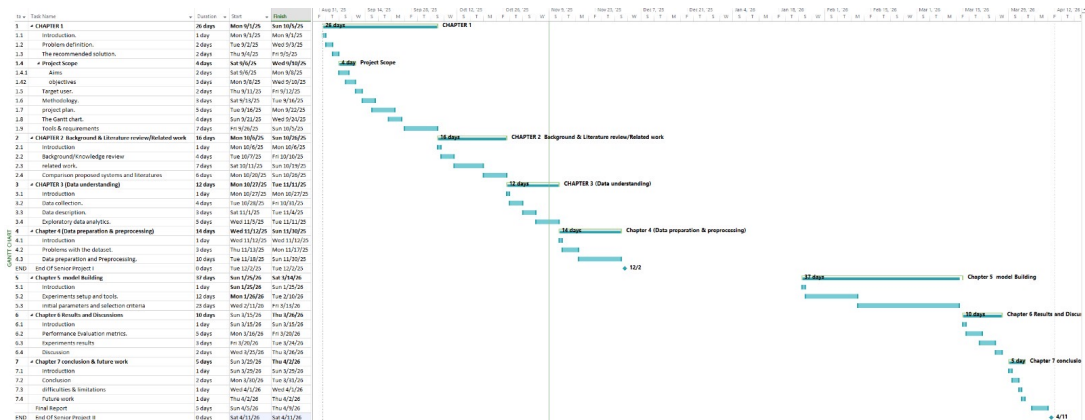


Figure 1

1.9 Tools and Requirements

Documentation: Overleaf – for writing and formatting project reports. (LaTeX)

Project Management: Microsoft Project – for creating and managing the Gantt chart.

Presentation: Microsoft PowerPoint – for preparing and presenting project deliverables.

Development Environment: Jupyter Notebook – for coding, data analysis, visualization, and experimentation.

Programming:

- **Language:** Python
- **Libraries:**
 - **pandas** – for data manipulation and analysis
 - **numpy** – for numerical computing
 - **matplotlib** – for plotting and visualization
 - **seaborn** – for enhanced statistical data visualization
 - **scikit-learn (sklearn)** – for preprocessing, evaluation metrics, and model assessment
 - **PyTorch** – for building and training deep learning models (Sequential, RNN, LSTM, GRU layers).

1.10 Conclusion

The proposed work focuses on enhancing predictive maintenance through robust, data-driven methodologies, primarily utilizing advanced deep learning models such as Deep LSTM, CNN-LSTM-Attention, or hybrid XGBoost/NN frameworks. By applying rigorous data preparation including feature engineering and improved RUL labeling methods on benchmark datasets like C-MAPSS, these models aim to deliver superior RUL prediction accuracy, resulting in minimized maintenance costs and significantly improved aircraft safety and reliability [9] [20].

2 Background & Literature review/Related work

2.1 Introduction

Accurate RUL estimation is a crucial task in Prognostics and Health Management (PHM) applications, particularly in the critical aviation sector [9]. RUL prediction helps engineers schedule maintenance to avoid catastrophic failures in both manufacturing and servicing sectors. Historically, maintenance has shifted from reactive and preventive approaches to the proactive approach of predictive maintenance (PdM) [4]. Given the complexity of modern aircraft systems and the difficulty of identifying potential failure mechanisms, data-driven methods have become the primary approach for RUL estimation, leveraging the wealth of data generated by advanced sensory technology [5].

2.2 Background (Knowledge review)

Maintenance techniques employed in industries are typically categorised into three types: reactive maintenance, preventive maintenance, and predictive maintenance (PdM) [2].

- **Reactive Maintenance:** Repair operation is considered only after a machine failure has occurred [2].
- **Preventive Maintenance:** Failure is prevented by performing maintenance operations based on time intervals, regardless of actual equipment condition. This approach often leads to unnecessary preventive maintenance actions [19].
- **Predictive Maintenance (PdM):** Estimates the time-to-failure of an equipment, allowing maintenance tasks to be scheduled proactively. PdM is also referred to as Condition-Based Maintenance (CBM) or PHM [9].

Turbofan jet engines are sophisticated, intricate, and highly precise pieces of engineering, regarded as the core of an aircraft's propulsion system. The manufacturing costs are high (ranging from \$50,000 to \$50 million), and overhaul costs can reach \$3 million. Critically, turbofan engines account for approximately 60% of the faults encountered in modern aircraft systems [10].

2.3 Related Work

The estimation of Remaining Useful Life (RUL) is a critical component of Prognostics and Health Management (PHM) in the aviation, allowing for optimal maintenance planning and enhanced safety [2] [30]. Due to the high complexity and dynamic nature of aircraft turbofan engines, identifying potential failure mechanisms through traditional model-based approaches is extremely difficult [5] [9]. Consequently, the field has largely shifted toward data-driven prognostic methods, leveraging vast sensor data generated in the Industry 4.0 era [5] [16] [22].

This related work section is structured thematically to provide an overview of the methodologies employed in RUL prediction. It begins with a review of traditional and foundational machine learning techniques, followed by a detailed synthesis of advanced deep learning architectures, specialized hybrid methods, and necessary data preprocessing strategies. Finally, the section identifies existing limitations that inform the direction of our current research.

2.3.1 Traditional and Model-Based Prognostics

Historically, RUL estimation relied on model-based methods, which establish degradation models using physics and statistics [9] [16].

- **Model-Based Techniques:** These methods typically employ algorithms like Kalman Filter (KF), Extended Kalman Filter (EKF), and particle filters to construct mathematical formulations of machine degradation based on multi-sensor time series data [2] [12]. This approach requires substantial expertise and domain knowledge, often leading to large prediction errors when faced with complex systems where the failure rate distribution function is difficult to model accurately [9] [15].
- **Statistical and Time Series Models:** Classical degradation analysis utilized statistical distribution models such as the Weibull distribution or the Eyring model [2] [15]. For forecasting failure rates over time, models like ARIMA (Autoregressive Integrated Moving Average) and Grey Verhulst models (suitable for handling less historical data or non-monotonic sequences) are employed, sometimes combined with other techniques like Complementary Ensemble Empirical Mode Decomposition (CEEMD) to improve the smoothness of the time series data [15].

2.3.2 Data-Driven Prognostics using Conventional Machine Learning (ML)

Data-driven approaches, especially supervised learning regression, analyze the relationship between sensor readings and RUL [10]. These methods are often tested as baseline models or integrated into hybrid architectures.

- **Regression Models:** Commonly adopted regression algorithms include Linear Regression (LR), Support Vector Regression (SVR), and Random Forest Regression (RFR) [10] [6].
 - SVR is noted for its ability to handle non-linear relationships in data, making it useful for precise failure prediction [1] [6].
 - RFR, as an ensemble method, showed strong early performance on the C-MAPSS dataset compared to other algorithms like K-Means and LR [10].
- **Neural Networks and Boosting:** The Multilayer Perceptron (MLP), a feedforward Artificial Neural Network, is widely used for modeling complex nonlinear relationships [6] [30]. Ensemble techniques like XGBoost (eXtreme Gradient Boosting) have also been successfully applied to estimate RUL, often showing performance superiority over single-model techniques [1].

2.3.3 Deep Learning (DL) Architectures for RUL Prediction

Deep Learning methods are favored for their ability to automatically learn features and capture temporal patterns in high-dimensional, sequential sensor data [16] [30].

- **Core Recurrent Architectures:** Long Short-Term Memory (LSTM) networks are prevalent in RUL estimation due to their efficacy in capturing long-term dependencies within the time series data, mitigating the vanishing gradient problem inherent in traditional RNNs [2] [5] [16] [9]. Bidirectional LSTMs (BiLSTM) further enhance this capability by processing sequences in both forward and backward directions, reducing information loss and improving temporal context capture [7] [20].

- **Convolutional Architectures:** Convolutional Neural Networks (CNNs), including 1D and 2D variants, are used primarily for feature extraction to capture key local spatial characteristics from the multivariate sensor data [16] [8] [9].
- **Hybrid Models:** Combining CNNs and RNN variants exploits the strengths of both. CNN-LSTM models are popular for feature extraction followed by temporal modeling [29] [8] [9]. The CNN-GRU architecture, combining CNNs with Gated Recurrent Units (GRU), is also applied, offering a balance of performance and greater training efficiency due to fewer parameters than LSTM [3].

2.3.4 Specialized Approaches: Ensemble, Transfer Learning, and Physics Integration

Recent work focuses on overcoming limitations such as data scarcity, complexity, and lack of interpretability through advanced frameworks.

- **Attention Mechanisms and Transformers:** Attention mechanisms are integrated into DL models (LSTM-Attention) to weigh the most relevant input features or time steps, thereby increasing prediction accuracy [5]. The Two-Stage Attention-Based Hierarchical Transformer (STAR) framework represents a novel step, incorporating a two-stage mechanism to capture both temporal attention and sensor-wise variable attention sequentially, demonstrating superior results on complex C-MAPSS subsets [29].
- **Ensemble Learning:** Methods often combine multiple diverse CNN models (1D and 2D) using a multistage ensemble learning strategy involving model selection techniques (APS, NNLS, OBS) and various weight learning methods to promote sparsity and improve generalization performance [21].
- **Transfer Learning and Meta-Learning:** These approaches address the challenge of data scarcity and domain shift when applying models to new equipment or operational conditions [16] [30]. Techniques like Domain Adaptation (DA) based on Maximum Mean Discrepancy (MMD) are integrated with multiscale CNNs to learn domain-invariant features with limited target data [16]. The Meta-Learning and Knowledge Discovery based Physics-Informed Neural Network (MKD-PINN) framework uses a first-order meta-learning strategy to enable rapid adaptation of model parameters with extremely few target domain samples (Few-Shot RUL Prediction) [30].
- **Physics-Informed Methods (PIML/PINN):** To integrate implicit physical knowledge, these frameworks embed known or discovered degradation dynamics into the neural network training. PIML can use stochastic methods (SDEs) to estimate governing physics models from noisy data (using K-means estimation for multi-modal data) and augment LSTM training with the learned mean and variance functions [18]. MKDPINN explicitly learns the non-linear Partial Differential Equations (PDEs) governing hidden state evolution and embeds them as a physical constraint into the PINN loss function [30].

2.3.5 Datasets, Pre-processing, and RUL Labeling

Effective data preparation is consistently emphasized as crucial for accurate RUL prediction [2].

2.3.6 Benchmark and Alternative Datasets

The vast majority of RUL prognostic studies, particularly those focused on turbofan engines, rely on the NASA Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) dataset.

- **C-MAPSS Characteristics:** This dataset contains simulated run-to-failure trajectories divided into four subsets (FD001–FD004), varying by the complexity of operational conditions (1 or 6) and fault modes (1 or 2) [29] [10].
- **Alternative Data Sources:** Some studies have focused on predicting non-RUL outcomes using different datasets:
 - **Failure Rate Prediction:** Research uses collected aircraft failure rate information (e.g., fuel control system failure rate over 38 months) to predict system health over time, utilizing models optimized for small time series data [15].

- **Failure Count/Rare Failure Prediction:** Studies utilize real-life historical maintenance data (pilot/maintenance reports) from specific airlines or large log-based datasets from the Aircraft Central Maintenance System (ACMS), often focused on identifying and predicting rare failures or estimating system failure counts over a period [6] [19]. The goal here shifts from remaining cycles to remaining time before a specific failure event occurs.
- **Synthetic Data:** Synthetic machine failure datasets are often generated to simulate realistic maintenance conditions and provide a robust foundation for model training, specifically addressing the real-world problem of insufficient failure data and class imbalance [28] [1].

2.3.7 Data Preprocessing and Feature Engineering

- **Feature Selection:** To manage high dimensionality, constant sensor measurements are discarded, typically resulting in 14 informative sensor readings being selected from the original 21 [30] [21]. Methods like Correlation Analysis are employed to select sensors demonstrating monotonous degradation behavior with RUL [2].
- **Normalization:** Scaling is essential to prevent models from assigning disproportionate importance to high-magnitude sensor readings. Both Standard Scaling (Z-score) and Min-Max normalization (scaling data to the range $[0, 1]$) are standard practices [25] [29].
- **RUL Target Labeling:** The standard is the Piecewise Linear (PwL) degradation model, which assumes RUL remains constant (clipped, $RUL_{max} = 125$ cycles) during the healthy phase and then decreases linearly until failure [9].
- **Advanced RUL Labeling:** To achieve a more realistic and engine-specific RUL label, advanced methods combine dimensionality reduction (via Autoencoders) and change-point detection mechanisms (using PwL fitting or logistic regression) to dynamically determine the point where degradation begins for each individual engine unit [20] [9].

2.3.8 Synthesis of Gaps and Alignment with Current Objectives

The existing literature highlights significant limitations that the current project must address to advance the reliability and practicality of RUL prediction:

1. **Idealized RUL Labeling:** A pervasive limitation is the generalized application of the Piecewise Linear (PwL) model with a single RUL maximum (125 cycles) across all engines, which *does not accurately reflect real-world engine degradation* [9]. The current project aims to adopt *engine-specific PwL target labeling* based on change point detection to achieve a more realistic representation of engine behavior [9].
2. **Handling Complexity (FD002/FD004):** Prediction accuracy tends to be significantly lower and noisier for subsets like FD002 and FD004, which involve *multiple operating conditions* and fault modes [2] [30] [29]. Models optimized for FD001 (single condition) often fail to generalize well. Our chosen architecture (e.g., using robust DL or hybrid models like BiLSTM or CNN-LSTM-Attention) must demonstrate proven robustness across all four complex subsets [8] [30] [20].
3. **Data Scarcity and Adaptability:** Deep learning models, including LSTMs, traditionally necessitate a *huge amount of labeled data* [5]. Techniques like *Few-Shot Learning*, *Transfer Learning*, or *Meta-Learning* are essential to ensure the prediction model can quickly adapt to new equipment or working conditions with limited samples [30].
4. **Interpretability vs. Performance:** Many high-performing hybrid and deep learning models *lack the ability to provide explanations* for their failure predictions, which is critical for enhancing *engineer trust and practical application* in regulated aviation environments [1] [29]. Future work should integrate Explainable AI (XAI) methods (like SHAP and LIME) to address this limitation.

Our project’s goal of developing an RUL model that enhances aircraft safety and efficiency will be achieved by leveraging advanced DL architectures (BiLSTM, CNN-LSTM, Attention) to tackle the inherent complexities of time-series sensor data, while explicitly addressing the identified gaps in RUL labeling and model generalizability.

2.4 Comparison between proposed systems and literatures

Performance evaluation is typically conducted using metrics such as Root Mean Squared Error (RMSE) and Score function. Lower RMSE and Score values indicate better performance.

Model/Approach	Dataset	Key Performance Metric	Comparison Insight	Source
Proposed ML (RFR/SVR/LR)	FD001	RMSE: 19.007, R^2 : 0.789	Outperformed RFR by Vimala Mathew (24.95 RMSE) and DL by Koen Peters (25.325 RMSE).	[10]
CNN-LSTM-Attention (Proposed)	FD001	RMSE: 15.977	Superior prediction accuracy compared to baseline CNN, LSTM, and CNN-LSTM models on C-MAPSS datasets.	[8]
1D-CNN-LSTM w/ Feature Eng. (Proposed)	FD001	RMSE: 16.1, Score: 437.2	Best RMSE among studies using only the test dataset (e.g., Aydemir and Acar's LSTM: 17.15 RMSE).	[9]
BiLSTM w/ Feat. Sel. & Change Point (Proposed)	FD001	RMSE: 18.71, % Error: 8.07%	Achieved 27.8% reduction in percentage error from the next best model (BiLSTM with change point only).	[20]
Hybrid XGBoost/NN (Proposed)	Synthetic Data	F1-Score (NN): 0.7199, PR AUC (XGBoost): 0.713	NN showed strong recall (0.9545), validating the hybrid model's robustness over baselines (LR, RFR, SVM).	[1]
DRL Approach (Proposed)	Turbofan Engines	Total Maintenance Cost: -36.3% (relative to corrective)	Reduced total maintenance cost by 29.3% compared to replacing engines at the mean-estimated-RUL, and prevented 95.6% of unscheduled replacements.	[13]
STAR Framework (Transformer-based) (Proposed)	FD002	RMSE: 13.47, Score: 784	Significantly outperformed state-of-the-art models for the challenging FD002 dataset (12% improvement in RMSE).	[29]

2.5 Conclusion

The literature confirms that RUL prediction for complex aircraft systems like turbofan engines is highly effective when using data-driven approaches, particularly deep learning models [2] [5]. Techniques such as LSTM networks, often integrated with CNNs and Attention mechanisms, demonstrate superior performance, especially when complemented by rigorous preprocessing steps like correlation analysis, data scaling, and the application of piecewise linear degradation models [8] [15]. The use of advanced methods like Deep Reinforcement Learning (DRL) further extends the application from pure RUL prediction to optimal maintenance scheduling, significantly reducing costs and unscheduled maintenance [13]. The consistent comparison against the NASA C-MAPSS dataset highlights the continual drive to reduce prediction errors (RMSE and Score), ensuring models are practical, reliable, and contribute substantially to aircraft safety and operational efficiency [2] [20].

3 Data Understanding

3.1 Introduction

Accurate Remaining Useful Life (RUL) prediction heavily relies on the quality and comprehensive understanding of the input data. For aircraft turbofan engines, this data is complex, high-dimensional, and often characterized by non-linearity and noise [30] [5]. This chapter details the characteristics of the data used for experimental testing, predominantly the widely adopted NASA C-MAPSS dataset [2], and outlines the initial stages of data collection, description, and exploratory analysis required before model development. Effective data preparation is essential for improving the accuracy of any predictive model, whether using Machine Learning (ML) or Deep Learning (DL) techniques [30] [12].

3.2 Data Collection

The primary dataset utilized for RUL prediction models is the Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) dataset [10] [13].

Source and Generation: The C-MAPSS dataset was developed by the National Aeronautics and Space Administration (NASA) in the MATLAB environment as a tool for simulating turbofan engines [5] [30]. The data consists of simulated run-to-failure trajectories for a small fleet of aircraft engines operating under realistic flight conditions [21] [25]. The original dataset was published in 2008 for the 1st International Conference on PHM [2].

Run-to-Failure Trajectories: The training data trajectories are composed of complete run-to-failure data [20], where faults grow in the system, providing specified RUL labels. The testing instances contain sensor measurements up to some moment prior to engine failure [8]. The objective is to predict the RUL cycle of engines in each sub-unit, with the actual RUL labels provided separately for validation.

Initial Conditions: Each engine trajectory in the dataset is unique, possessing distinct manufacturing variation and initial wear, which are unknown to the researcher. All engines start operating normally, with faults originating sometime later in their life cycle [20].

While C-MAPSS is standard for RUL regression, other studies utilize different datasets for specific failure prediction or reliability analysis tasks:

N-CMAPSS: A newer dataset, N-CMAPSS, provides the full history of engine trajectories, starting from a healthy condition until failure occurs [22]. It simulates 90 flights divided into three classes based on flight length (1–3 hours, 3–5 hours and, 5 hours) [22].

Log-Based/Maintenance Data: Real-life data, often in the form of pilot and maintenance reports of faults/failures from operational aircraft, is used for reliability analysis, especially when the overall sample size is small [19]. Large log-based datasets derived from the aircraft central maintenance system (ACMS), comprising error messages, have been used to identify and predict rare failures [7].

Synthetic Data: Synthetic data is sometimes generated to simulate realistic equipment failure scenarios to provide a robust foundation for model training, particularly addressing issues of class imbalance and data control [1].

3.3 Data description

C-MAPSS Data Composition

The C-MAPSS dataset is composed of multivariate time series data describing engine operation and degradation:

Variables and Parameters: The model simulates engine degradation using fourteen input parameters related to five rotating components of the engine. The data comprises three operational settings (Setting_1, Setting_2, and Throttle Resolver Angle (TRA)) and 21 sensor readings [8] [2].

Sensor Variety: The 21 sensors collect data on various engine parameters, including temperature, pressure, shaft speed, fuel flow, and bypass ratio [10].

Engine Configuration: The simulation model includes five modules: fan, low-pressure turbine (LPT), high-pressure turbine (HPT), low-pressure compressor (LPC), and high-pressure compressor (HPC), all of which can be affected by the degradation process [29].

Subsets (FD001–FD004)

The C-MAPSS sensor trajectories are divided into four distinct sub-datasets—FD001, FD002, FD003, and FD004—which correspond to different operational conditions and fault modes.

Dataset	Operating Conditions	Fault Modes	Training Engines	Test Engines	Training Rows	Test Rows
FD001	1 (Single)	1	100	100	20631	13096
FD002	6 (Multiple)	1	260	259	53759	33991
FD003	1 (Single)	2	100	100	24720	16596
FD004	6 (Multiple)	2	248	249	61249	41214

Table 1C-MAPSS Dataset Subsets with Number of Rows

- Datasets FD002 and FD004 are more complex as they involve six distinct operating conditions and contain twice as many training instances as FD001 and FD003 [3].
- The fault modes include fan degradation and High-Pressure Compressor (HPC) blades degradation [10].
- The total sample length of the training sets for FD001–FD004 varies, ranging from 20,630 to 61,248 total cycles [30].

Data Characteristics

Sensor readings in the C-MAPSS dataset exhibit several key characteristics:

Noise Contamination: The sensor measurements are polluted by sensor noise [9].

Irregular Behavior: Subsets FD002 and FD004, having 6 operating conditions, possess irregular behavior, making RUL prediction for these complex sequences a challenge [2].

Magnitude Variation: Sensor readings vary greatly in magnitude. For example, some sensors are in the 1400 range, while others may have values less than 1, requiring normalization to prevent bias during training [10].

Table of Engine Sensor Parameters

Sensor	Parameter	Description with units	Trend
1	T2	Total Temperature in fan inlet (oR)	~
2	T24	Total Temperature at LPC outlet (oR)	I
3	T30	Total Temperature at HPC outlet (oR)	I
4	T50	Total Temperature at LPT outlet (oR)	I
5	P2	Pressure at fan inlet (psia)	~
6	P15	Total pressure in bypass-duct (psia)	~
7	P30	Total pressure at HPC outlet (psia)	D
8	Nf	Physical fan speed (rpm)	I
9	Nc	Physical core speed (rpm)	I
10	Epr	Engine pressure ratio (—)	~
11	Ps30	Static pressure at HPC outlet (psia)	D
12	Phi	Ratio of fuel flow to Ps30 (psi)	I
13	NRf	Corrected fan speed (rpm)	I
14	Nrc	Corrected core speed (rpm)	D
15	BPR	Bypass ratio (—)	I
16	farB	Burner fuel air ratio (—)	~
17	htBleed	Bleed enthalpy (—)	I
18	NF-dmd	Demanded fan speed (rpm)	~
19	PCNR-dmd	Demanded corrected fan speed (rpm)	D
20	W31	HPT coolant bleed (lbm/s)	D
21	W32	LPT coolant bleed (lbm/s)	D

3.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) serves as a critical step to understand the dataset's structure, identify degradation patterns, and prepare features for Remaining Useful Life (RUL) modeling. The process involves deriving the RUL target, examining sensor behavior, removing irrelevant features, and ensuring data consistency through scaling and normalization [29].

RUL Calculation

The target feature, Remaining Useful Life (RUL), must be calculated for each engine cycle.

- **Basic Formula:** RUL is calculated by finding the difference between the maximum cycle count achieved by a specific engine and the current time cycle:

$$RUL = max_cycle - time_cycle$$

Here, *max_cycle* is the last *time_cycle* for that particular engine unit.

The necessity for applying a piecewise linear degradation function stems from the observation that in real-world operations, RUL is fundamentally a linear decreasing function with respect to time as the system's efficiency degrades; however, no degradation is present in the sensor readings when the system begins operation. Consequently, during the initial stages of operation, engine degradation is typically negligible [29].

To create RUL target labels that accurately reflect realistic degradation behavior and to improve prediction accuracy, a piecewise linear degradation model (or function) is applied during the preprocessing phase [8]. This model divides the engine's operational life into two distinct stages:

1. **Constant Phase (Healthy Stage):** During this initial period, which is characterized by normal operating conditions, the RUL label is artificially clipped and remains constant. This initial RUL value is typically set to a fixed maximum, such as 125 cycles [16], and represents the system's phase of minimal degradation.
2. **Degradation Phase (Linear Decrease):** The RUL begins to decrease linearly only after the deterioration point, also known as the inflection point. This point indicates a noticeable increase in sensor trends and marks the transition into the linear degradation stage [9]. Following this point, the remaining RUL labels are represented as a linear line that decreases until the point of zero life cycle time.

This truncated linear model is crucial because the accurate calculation of this starting point of degradation (initial RUL) has a great impact in determining accurate RUL predictions, ensuring the model focuses its learning on the true degradation region of the equipment [30].

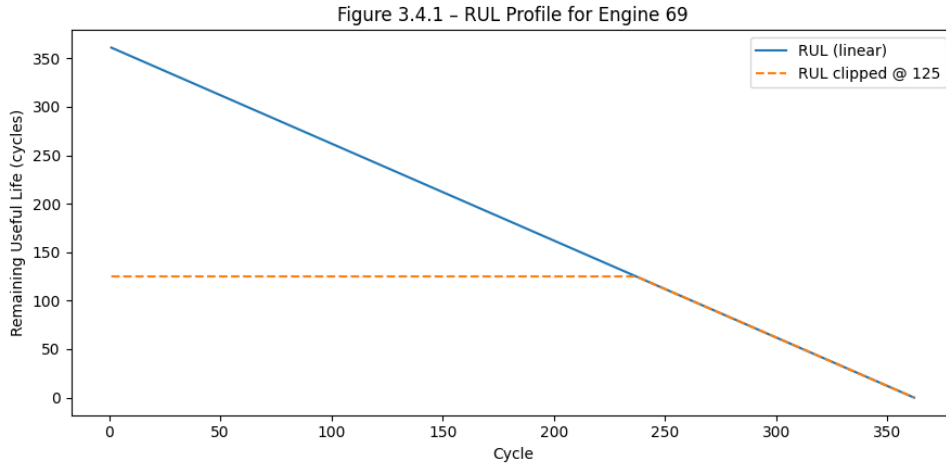


Figure 3.4.1: Remaining Useful Life (RUL) Profile for a Single Engine

Feature Analysis and Selection

The analysis of sensor data trends, known as Feature Analysis and Selection, is a crucial preliminary step in Exploratory Data Analysis (EDA) for Remaining Useful Life (RUL) prediction [6]. This systematic identification of sensors that provide meaningful degradation information helps in dimension reduction, improving the accuracy of models, reducing computational time, and ensuring that models are trained on reliable data [20].

The C-MAPSS dataset, which includes 21 sensor readings and 3 operational settings, is the primary focus for this analysis.

Sensor Trend Identification and Categorization

To determine the utility of each sensor, their readings are plotted against the operational cycle or the computed RUL to analyze their degradation pattern with respect to time [9]. This process reveals three main categories of sensor behaviour:

1. **Monotonous Trends (Used for Prediction):** These sensors exhibit a clear, systematic change (increasing or decreasing) over time, indicating degradation [9]. These are the most trendable sensors values and are selected to create the health indicator (HI).
 - **Increasing Trends (Positive/Ascending):** These parameters increase with time. For the FD001 subset of the C-MAPSS dataset, sensors demonstrating an ascending trend include: 2, 3, 4, 8, 9, 11, 13, 15, and 17 [10].
 - **Decreasing Trends (Negative/Descending):** These parameters decrease with time. For FD001, sensors demonstrating a descending trend include: 7, 12, 14, 20, and 21.
2. **Constant or Irregular Trends (Discarded):** These sensor readings remain constant, have zero correlation with RUL, or exhibit complex, oscillating behavior that offers little predictive value. Sensors with constant values are discarded because they do not provide valuable information for RUL prediction and increase computational resource usage.
 - For the FD001 subset, sensors identified as having constant or random trends are typically 1, 5, 6, 10, 16, 18, and 19. Sensor 6 is explicitly noted as having a constant value in FD001. The sensor readings abandoned due to irregular/unchanged output include 1, 5, 6, 10, 16, 18, 19, and 22 [18].

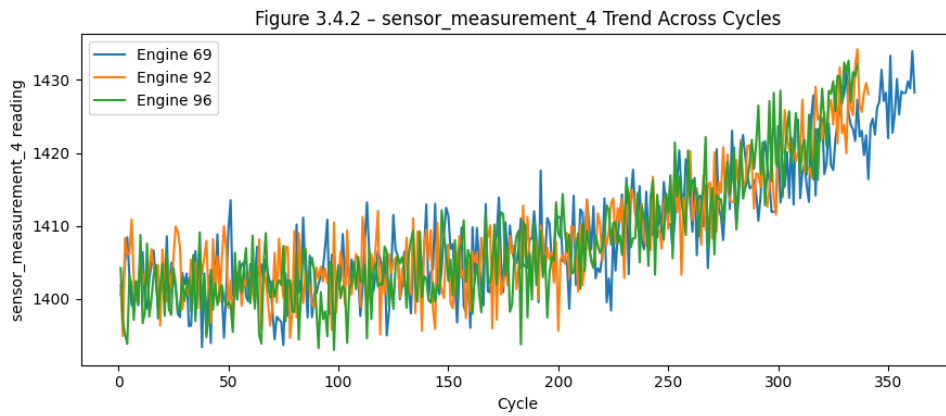


Figure 3.4.2a: Sensor 4 Trend Across Cycles (Gradual Degradation)

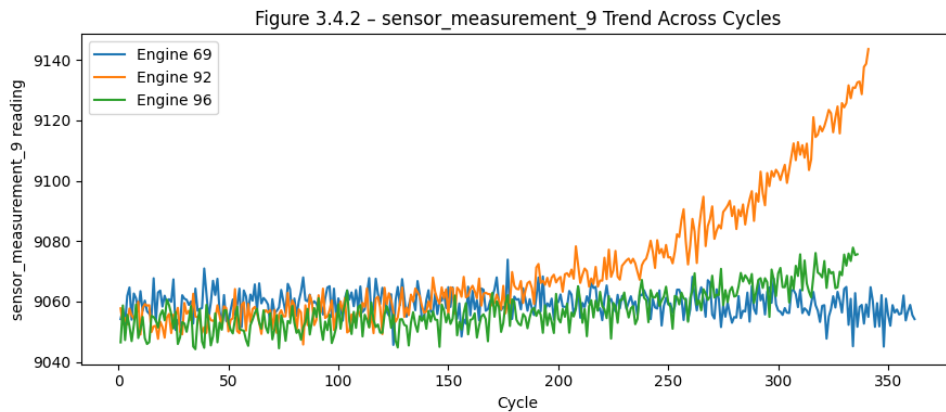


Figure 3.4.2b: Sensor 9 Trend Across Cycles (Rapid Degradation)

Methods for Feature Selection

To formalize the selection process and achieve dimensionality reduction, researchers employ specific methodologies

Manual and Visual Selection: Initial feature selection is performed manually by analyzing the importance of each feature and plotting sensor data to determine the trend.

Correlation Analysis: This technique is a standard preprocessing step used to discover the relevance of features with RUL. The correlation coefficient (r) is computed to assess the relationship between sensors and RUL labels. This algorithm excludes sensor values that have a very little or zero correlation with RUL. Based on this analysis, the remaining 14 useful sensor measurements are selected as input for the prediction model [25].

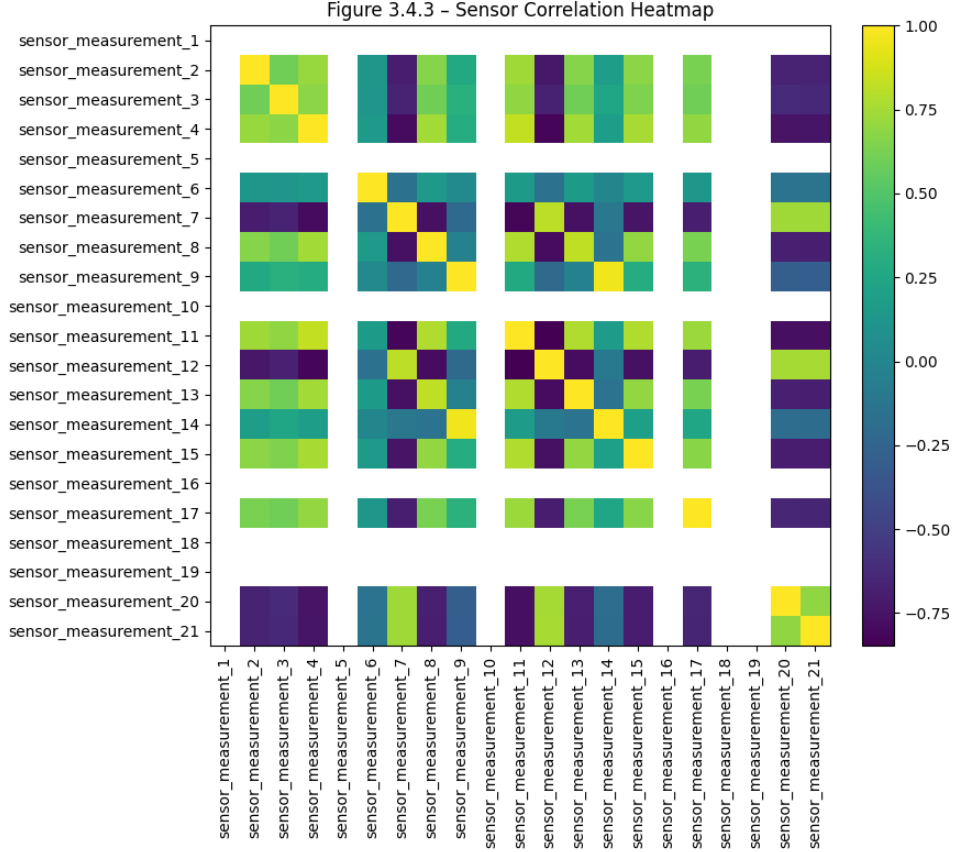


Figure 3.4.3: Correlation Heatmap of Sensor Measurements)

Random Forest (RF) and ReliefF: Algorithms like Random Forest are used to evaluate the importance ranking of each monitoring parameter and eliminate features with the lowest weight score. The ReliefF algorithm helps identify the most and least influential parameters, which is particularly useful in datasets designed for predicting failure counts [4].

Data Scaling and Normalization

The necessity for Data Scaling and Normalization is a consistently highlighted preprocessing step in RUL prediction research, as sensor readings often vary significantly in their scales and physical units, ranging from tens to thousands [22]. Using these raw values directly for training networks would cause the accuracy to drop significantly or lead to model bias, where disproportionate importance is given to high-magnitude sensor readings (e.g., sensor values in the 1400 range versus those less than 1). Normalization is therefore necessary to standardize the data, bring the values into a fixed range, and ensure an equal contribution from each feature vector for efficient training of the network [25].

The sources detail the application and choice of two primary scaling techniques:

Min-Max Normalization

- **Process:** Min-Max normalization (or MinMax scaling) is a method used to map the values of the data into a fixed range, typically between $[0, 1]$. The process involves subtracting the minimum value and dividing by the range of values within a feature.
- **Application:** This technique is utilized to adjust feature columns to have a similar scale, which is crucial when features vary significantly in their scales. Min-Max scaling is valuable in scenarios where maintaining the original data distribution is not critical, and consistent feature scaling is more crucial. The LSTM+A method and its variants often focus on keeping the outputs and inputs within the $[0, 1]$ range for stabilization purposes of learning.

Z-Score Standardization

- **Process:** Z-Score normalization (or Standard Scaling) is used to transform features into standardized variables. This involves calculating the Z-score by subtracting the mean (μ) of each feature vector and dividing by its standard deviation (σ). This process results in standardized features with zero mean and unit variance.
- **Application:** This scaling method is used to bring the values into a fixed range. Standard Scaling was explicitly utilized in hybrid Machine Learning (ML) models like Linear Regression, SVR, and Random Forest Regression on the FD001 dataset.

Condition-Based Standardization (CS)

- **Process:** When dealing with datasets that feature multiple operational conditions, such as the C-MAPSS subsets FD002 and FD004, a more complex approach is often required due to signal fluctuations caused by switching operating conditions. Condition-Based Standardization (CS), as proposed in some literature, addresses this by dividing the data into different operating condition groups according to operational parameters. Standardization calculations are then performed independently within each operating condition group. The normalization formula for CS typically includes the minimum (min) and maximum (max) measurements of a sensor (s) under a specific operating condition (o).
- **Goal:** This strategy is critical because it effectively filters out signal fluctuations caused by operating condition switching and allows the model to more clearly reveal the intrinsic degradation patterns related to equipment health status. The resultant signal often presents smoother and more trend-oriented degradation characteristics compared to global standardization, where degradation trends are largely submerged by dramatic operational condition fluctuations.

Overall, normalization is a widely used preprocessing step, ensuring that the models receive filtered signals that have the same degree of range for efficient training of the network [22].

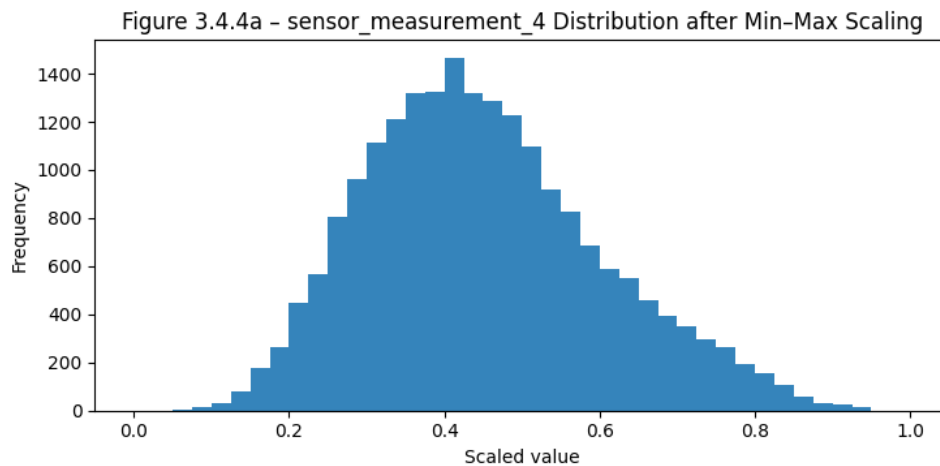


Figure 3.4.4: Sensor Value Distributions After Normalization and Condition-Based Standardization)

3.5 Conclusion

The foundation of the proposed RUL prediction models is the C-MAPSS benchmark dataset, which contains complex, multi-source, non-linear time-series data crucial for simulating turbofan engine degradation. Initial data exploration, particularly for identifying relevant sensors and understanding the degradation process, is necessary before moving to the detailed data preparation and preprocessing phase [13].

4 Data preparation & preprocessing

4.1 Introduction

The accurate prediction of Remaining Useful Life (RUL) using data-driven methods relies fundamentally on providing the machine learning or deep learning models with clean, meaningful, and optimally structured data [5]. Data preparation is considered the core module of machine learning and the decision-making system [5] [4]. It manages the raw input data to make it useful for decision-making and forecasting future events.

Given the complex nature of turbofan engine sensor readings, effective preprocessing steps are crucial for improving the prediction accuracy of deep learning networks, such as Long Short-Term Memory networks [2]. Preprocessing typically includes data cleaning, feature engineering, and data transformation [12] [8].

4.2 Problems with the dataset

The datasets used for RUL prediction, particularly the benchmark C-MAPSS data, present several challenges that must be addressed during preprocessing:

1. **Noise and Inconsistency:** Sensor readings in datasets like C-MAPSS and real-world maintenance logs are subject to **sensor noise** and may contain **inconsistent or irrelevant data** [9]. For example, the C-MAPSS training dataset's sensor data are contaminated with a fair amount of noise, and each trajectory has its own initial wear pattern [25]. Training ML algorithms with noisy or irrelevant data negatively affects prediction performance and complexity [6].
2. **High Dimensionality:** The C-MAPSS dataset consists of **21 sensor readings** plus operational settings [20]. This high dimensionality complicates RUL prediction problems and increases **computational requirements** [2]. Feature selection is needed to manage the large number of monitoring parameters [30].
3. **Varying Scales and Units:** The magnitude of different sensor readings varies greatly, ranging **from tens to thousands** [20]. For instance, values of sensor 3 and 4 may be in the **1400 range**, while sensor 16 values are **less than 1** [29]. Using these raw values for training can cause **accuracy to drop significantly** and result in unequal importance being given to higher-magnitude sensors [8].
4. **Complex Degradation Trends:** Datasets like FD002 and FD004 are more complex as they involve **six operating conditions and multiple fault modes**, leading to **irregular sensor behavior**. This complexity can mask the **fundamental degradation trend**, making RUL prediction a challenge [5].
5. **RUL Target Labeling Distortion:** In the initial stages of operation, engines are healthy, meaning degradation is negligible. Using a strictly linear decrease in RUL from the start **does not accurately reflect the real world** and creates **label noise** in early cycles [21]. Furthermore, using a **common RUL max** for all engines limits the reflection of realistic degradation behavior, which differs for each engine unit due to varying initial wear [20].
6. **Class Imbalance (for Classification/Failure Detection):** In failure forecasting tasks, the dataset often suffers from **class imbalance** [4], where non-failure instances significantly outnumber failure events (approximately **5% of records represent failure events** in a synthetic dataset). Training with imbalanced data degrades model performance and causes unreliable prognostics.

4.3 Data preparation and Preprocessing

Preprocessing is implemented to overcome data shortcomings and includes steps such as feature selection, RUL target labeling, data cleaning, and normalization.

4.3.1 Feature Selection and Reduction

The objective of feature selection is to remove irrelevant and noisy data, selecting a minimal subset of features necessary to achieve maximum accuracy and reduce model complexity.

- **Removal of Constant/Irrelevant Sensors:** Sensors that **do not alter over time** or remain constant throughout the engine's lifetime are typically **discarded** because they do not provide valuable information for RUL prediction. For the C-MAPSS dataset, this process often reduces the number of sensors from 21 to **14 informative sensor measurements**.

- **Trend Identification:** Visual analysis or automated methods categorize sensors based on their behavior relative to RUL or time cycle:
 - **Monotonous/Trending Sensors** (Ascending or Descending) are selected for prediction (sensors 2, 3, 4, 7, 8, 9, 11, 12, 13, 14, 15, 17, 20, 21 in FD001).
- **Statistical Selection:**
 - **Correlation Analysis** is performed to choose sensors that possess a monotonous behavior or high strength of correlation with the RUL output label, discarding those with less than 10% correlation.
 - The **Random Forest algorithm** is used to generate an importance ranking value for monitoring parameters to eliminate features with low importance.
 - The **ReliefF algorithm** is used to find the most effective parameters in maintenance data for failure count prediction.
- **Dimensionality Reduction:** Techniques like the Autoencoder are used to perform dimension reduction by extracting a single encoded signal from multiple sensor readings, which facilitates change point detection and simplified analysis. PCA (Principal Component Analysis) is also a feature extraction process used for dimensionality reduction.

4.3.2 RUL Target Labeling

RUL target labels must be calculated accurately from the raw data.

- **Piecewise Linear (PwL) Degradation Function:** This is the most common method for target RUL construction. The PwL model assumes RUL is constant during the healthy phase (clipping the RUL) and decreases linearly after a certain degradation point. This is done to reflect the non-linear reality that RUL does not decrease immediately when the engine is new.
- **Improved/Engine-Specific PwL:** To achieve greater accuracy, an improved piecewise linear degradation model is implemented to determine the starting point of deterioration (initial RUL) for each engine unit. The initial RUL value is calculated using algorithms that analyze the rate of rise in sensor data. This is critical as the exact starting point of degradation has a great impact on prediction accuracy.
- **Change Point Detection:** Specific mechanisms, such as PwL fitting or techniques that leverage a dimensionality-reduced signal, are used to determine the exact breakpoint (change point) where linear degradation begins for engine-specific labeling.

4.3.3 Data Scaling and Normalization

Normalization adjusts features to have a similar scale, which is essential because the large differences in dimensions and orders of magnitude of indicators can affect model prediction and convergence speed.

- **Min-Max Normalization:** This method scales and transforms data to fit within a specified range, typically between 0 and 1. It is applied to sensor measurements per sensor to ensure comparability and eliminate bias.
- **Z-Score Standardization (Standard Scaling):** This technique calculates the Z-score of each feature by subtracting the mean (μ) and dividing by the standard deviation (σ). This results in standardized features with zero mean and unit variance. It is commonly used in studies applying ML algorithms and sometimes for DL models.
- **Condition-Based Standardization (CS):** For complex datasets with multiple operating conditions (FD002, FD004), the CS method is utilized. CS first divides the data into groups based on operating parameters and then performs standardization independently within each group. This effectively filters out signal fluctuations caused by operating condition switching and reveals cleaner degradation trends.

4.3.4 Data Filtering and Imbalance Handling

- **Filtering/Smoothing:** Raw sensor data are often filtered to reduce noise interference and uncover the main degradation trend. Techniques include the Moving Median filter and the Single Exponent Smoothing (SES) algorithm.
- **Noise Elimination (Clustering):** A modified K-means algorithm is used to eliminate noisy and inconsistent data from maintenance records. In some DL work, unsupervised clustering (e.g., K-means) is used to determine operating regimes or create labels for unsupervised preprocessing.
- **Handling Imbalance:** For classification tasks where failure events are rare, the Synthetic Minority Over-sampling Technique (SMOTE) is employed to rebalance the dataset by creating synthetic instances of the minority (failure) class.

4.3.5 Advanced Feature Engineering

Beyond basic selection and scaling, new features may be created to enhance model performance:

- **Difference Feature Construction:** A handcrafted feature is constructed based on the difference between the sensor measurements in every cycle and the first sensor measurement (or an initial measurement). This new feature is added to the dataset and helps improve RUL prediction in the middle- and early-life stages.
- **Polynomial Features:** Polynomial feature engineering is utilized to increase the number of features and enhance their importance, particularly when using traditional ML models.
- **Sliding Time Window:** For sequential DL models (LSTM, CNN), data must be prepared in the correct format. The sliding time window technique is used to generate input samples that capture the history of sensor readings over a defined time length (L_w), which serves as the input tensor for the network.

4.3.6 Application of the Large Language Model (LLM) as a Coding Assistant

To ensure the highest levels of accuracy and efficiency in implementing the preprocessing algorithms, a Large Language Model (LLM) specifically Gemini was strategically employed as an intelligent coding assistant. This approach provided substantial benefits compared to traditional manual coding:

- **Automation of Complex, Domain-Specific Tasks:** The LLM converted complex methodological requirements, such as the Piecewise Linear RUL calculation (clipped at a maximum of 125 cycles) and the structuring of data into 3D sequences for LSTM/GRU model requirements—into executable Python code. This significantly reduced the time spent on manual code development and testing.
- **Ensuring Best Practices Compliance:** Utilizing the LLM guaranteed that the resulting code adhered to established best practices within RUL research (correct scaling rules using StandardScaler and the exclusion of zero-variance features), thereby enhancing the reliability of the implementation.
- **Accelerated Workflow and Shift in Focus:** The ability to generate accurate code rapidly accelerated the entire preprocessing phase from days to minutes. This crucial time saving allowed us to shift focus from debugging to more critical analytical and methodological considerations.

Table 2 Comparison of Traditional vs. LLM-Assisted Preprocessing Workflows

Metric	Traditional Workflow (Manual Coding)	LLM-Assisted Workflow (Gemini)	Impact on Research
Time to Develop/Debug	≈ 4 hours	≈ 30 minutes	87.5% reduction in development time, enabling a faster transition to model architecture design.
Code Reliability (Initial Run)	Initial errors often included indexing errors and incorrect application of the scaler (data leakage).	Near 100% success on first run for complex functions (e.g., <code>create_sequences</code>).	Reduced time spent on debugging and validation, increasing confidence in the data preparation quality.
Code Complexity Management	Required complex nested loops and manual array reshaping using NumPy.	Code was automatically generated, optimized, and clearly commented by the LLM.	Allows the researcher to focus on the methodological concept rather than the coding syntax.
Target Implementation	Required manual lookup and implementation of the Piecewise Linear RUL clipping condition ($\min(\text{RUL}, 125)$).	Condition was accurately and automatically integrated based on the high-level prompt.	Ensures immediate adherence to re-established RUL best practices.

Conclusion on Workflow Efficiency The use of the LLM was not merely a replacement for programming; it served as a productivity multiplier. By automating the transformation of complex methodological requirements—such as the specific RUL clipping logic and the essential 3D sequence structuring for the LSTM/GRU input—the LLM allowed us to shift focus entirely from low-level code implementation to high-level model evaluation and analysis. This approach significantly enhances the reproducibility and speed of the overall RUL prediction pipeline.

4.4 Conclusion

Data preparation and preprocessing are integral, non-negotiable phases in RUL prognostics, acting as a crucial module that ensures successful model training and optimal performance. The process systematically addresses key dataset problems, including high dimensionality (via correlation analysis and sensor trend checks), magnitude variation (via Min-Max or Z-Score normalization), and the inaccuracy of RUL labels (via piecewise linear degradation clipping and change point detection). Furthermore, specialized techniques like Condition-Based Standardization (CS) are employed to isolate degradation trends from operational noise in complex multi-condition datasets. The output of these comprehensive preprocessing steps—which includes carefully selected, scaled, and re-labeled data—is directly input into the prognostic models, significantly influencing their ability to attain minimum prediction errors.

5 Model Building

5.1 Introduction

This chapter presents the implementation and training process of the proposed Remaining Useful Life (RUL) prediction framework using the NASA C-MAPSS turbofan engine dataset. After completing the preprocessing and feature engineering stages, several Machine Learning (ML) and Deep Learning (DL) models were developed and evaluated to predict engine degradation behavior and estimate the remaining operational cycles before failure.

The experiments were conducted separately on the four standard C-MAPSS subsets (FD001, FD002, FD003, and FD004) to evaluate the proposed models under different operational conditions and fault modes. These subsets vary in complexity, allowing a more comprehensive analysis of model robustness and generalization capability.

The implemented models include:

- Bidirectional Long Short-Term Memory (BiLSTM)
- GRU with Attention
- XGBoost
- Ensemble Learning

The goal of this chapter is to describe the experimental setup, model architectures, training procedures, and evaluation strategies used throughout the project.

5.2 Experimental Setup

All experiments were implemented using Python within the Jupyter Notebook environment. Multiple scientific computing and machine learning libraries were utilized during development and evaluation.

- NumPy and Pandas were used for data processing and manipulation.
- Matplotlib and Seaborn were used for visualization.
- Scikit-learn was used for preprocessing, evaluation metrics, and cross-validation.
- PyTorch was used for implementing deep learning models.
- XGBoost was used for gradient boosting regression.

The experiments were executed on the NASA C-MAPSS dataset, which contains multivariate time-series sensor measurements collected from simulated turbofan engines operating until failure.

5.3 Dataset Configuration

The experiments were performed separately on the four C-MAPSS subsets:

Table 3C-MAPSS Dataset Subsets

Subset	Operating Conditions	Fault Modes
FD001	Single	Single
FD002	Multiple	Single
FD003	Single	Multiple
FD004	Multiple	Multiple

FD001 is considered the simplest subset, while FD004 is the most challenging due to multiple operating conditions and fault modes.

5.4 Data Preparation and Sequence Generation

Several preprocessing steps were applied before model training. Constant and low-correlation sensors were removed to reduce unnecessary information and improve model efficiency.

The remaining features were normalized using StandardScaler to improve optimization stability during training.

For deep learning models, a sliding window approach was used to generate sequential input samples. Each sequence contained historical sensor readings from previous operational cycles, allowing the recurrent models to learn temporal degradation behavior.

The sequence length was experimentally set to 30 cycles.

Although Min-Max normalization was examined during the exploratory phase (Section ??), StandardScaler (Z-score normalization) was selected for model training. This choice is motivated by three factors: (i) Adam-based optimization converges more reliably when input features have approximately zero mean and unit variance; (ii) the piecewise-linear RUL target is unbounded above, which is more compatible with standardization than bounded scaling; and (iii) StandardScaler is less sensitive to sensor-level outliers arising from sensor noise in the C-MAPSS data. The scaler was fitted exclusively on the training partition of each split, with no statistics from the validation or test sets entering the scaling parameters.

5.5 RUL Labeling Strategy

The Remaining Useful Life target values were generated using a Piecewise Linear degradation function.

To reduce instability during healthy operational stages, the RUL values were clipped at a maximum threshold of 125 cycles. After the degradation point, the RUL values decrease linearly until engine failure occurs.

The clipping strategy improves training stability and reduces the influence of excessively large target values during early engine cycles.

5.6 BiLSTM Model

The Bidirectional Long Short-Term Memory (BiLSTM) model captures temporal dependencies in both forward and backward directions across the 30-cycle input window.

The architecture consists of two stacked Bidirectional LSTM layers, each with a hidden size of 64 per direction (128 units per time step). An inter-layer dropout rate of 0.2 is applied between the two LSTM layers. The hidden state of the final time step is passed to a regression head composed of a fully connected layer ($128 \rightarrow 64$), a ReLU activation, a dropout layer (rate = 0.2), and a final linear layer ($64 \rightarrow 1$) that produces the scalar RUL estimate.

Bidirectional processing allows the network to integrate context from both past and future cycles within each window, which has been shown to outperform unidirectional LSTM for fixed-window sequence regression tasks. The BiLSTM architecture consisted of two bidirectional LSTM layers with 64 hidden units and dropout rate of 0.2 followed by fully connected dense layers for regression output.

5.7 GRU with Attention Model

The GRU with Attention architecture combines Gated Recurrent Units (GRU) with an attention mechanism.

The GRU component captures sequential degradation behavior, while the attention mechanism allows the model to focus on the most informative temporal patterns during prediction.

This improves learning efficiency and enhances the model’s ability to identify critical degradation signals.

The architecture consists of a single GRU layer with 64 hidden units, an additive (Bahdanau-style) attention mechanism applied over all 30 timesteps of the hidden-state sequence, followed by a dense regression head: $\text{Linear}(64 \rightarrow 64) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.2) \rightarrow \text{Linear}(64 \rightarrow 1)$. The attention weights $\alpha_t = \text{softmax}(v^\top \tanh(Wh_t))$ are used to form a context vector $c = \sum_{t=1}^{30} \alpha_t h_t$, which is then passed to the regression head.

5.8 XGBoost Model

XGBoost was implemented as a machine learning baseline for comparison against the deep learning models. XGBoost is a tree-based gradient boosting algorithm that learns non-linear relationships between features and the target through an additive ensemble of decision trees.

Since XGBoost does not natively process temporal sequences, the input was constructed by extracting only the *last time step* of each 30-cycle window. This yields a tabular feature vector per sample that captures the most recent sensor state immediately before each RUL prediction point.

The XGBoost regressor was trained with the squared-error objective and the following hyperparameters: learning rate $\eta = 0.03$, maximum tree depth = 6, row subsample ratio = 0.8, column subsample per tree = 0.8, L2 regularization $\lambda = 1.0$, and a maximum of 5,000 boosting rounds with early stopping (patience = 100 rounds) on the validation RMSE.

Despite lacking explicit temporal modeling capability, XGBoost achieves competitive performance on several subsets and provides architectural diversity to the ensemble.

5.9 Ensemble Learning

An Ensemble model was developed by combining predictions from BiLSTM, GRU-Attention, and XGBoost.

The ensemble approach aims to utilize the strengths of different architectures:

- Deep learning models capture temporal degradation behavior.
- XGBoost captures feature-based relationships.
- Ensemble averaging reduces prediction variance and improves robustness.

The final ensemble prediction is computed as the simple arithmetic mean of the three model outputs:

$$\widehat{\text{RUL}}_{\text{ensemble}} = \frac{1}{3} \widehat{\text{RUL}}_{\text{BiLSTM}} + \frac{1}{3} \widehat{\text{RUL}}_{\text{GRU+Attn}} + \frac{1}{3} \widehat{\text{RUL}}_{\text{XGBoost}}$$

where $\widehat{\text{RUL}}_{\text{BiLSTM}}$, $\widehat{\text{RUL}}_{\text{GRU+Attn}}$, and $\widehat{\text{RUL}}_{\text{XGBoost}}$ are the per-sample predictions of the three constituent models. Equal weights (1/3 each) were assigned because no significant performance advantage was observed for tuned weighting on the validation set.

5.10 Training Configuration

Several hyperparameters were experimentally selected to improve convergence and model performance.

Table 4 Model Hyperparameters

Parameter	Value
Sequence Length	30
Batch Size	128
Learning Rate	0.001
Epochs	20
Optimizer	Adam
Dropout Rate	0.2
Hidden Units	64

Training objective. The recurrent models (BiLSTM and GRU with Attention) were trained using the Huber loss with $\delta = 10$. Huber loss behaves as MSE for small residuals ($|r| \leq \delta$) and as MAE for large residuals; this moderates the gradient contribution of the clipped healthy-phase samples, where label residuals are systematically large, and produces more stable optimization than pure MSE for RUL prediction with piecewise-linear targets.

Gradient stabilization. Gradient clipping with a maximum L2 norm of 1.0 was applied at each optimization step to prevent gradient explosion in the recurrent layers.

Model selection. Rather than conventional early stopping, model checkpointing was used: at the end of each epoch, validation MAE was evaluated and the parameter set producing the lowest validation MAE so far was stored. Training ran for the full epoch budget, and the best checkpoint was restored before final test evaluation. This ensures the reported test metrics correspond to the best validation-time configuration, not necessarily the final epoch.

5.11 Cross Validation

To evaluate model robustness and stability, Group K-Fold Cross Validation was implemented.

Unlike traditional K-Fold splitting, Group K-Fold ensures that engine units appearing in the training set do not appear simultaneously in the validation set, preventing data leakage between engine trajectories.

Cross-validation experiments demonstrated stable model performance across different folds and improved the reliability of the evaluation process.

5.12 Evaluation Metrics

The implemented models were evaluated using:

- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)

RMSE was selected as the primary evaluation metric because it penalizes large prediction errors more heavily, making it highly suitable for predictive maintenance applications.

5.13 Conclusion

This chapter presented the implementation of multiple machine learning and deep learning models for Remaining Useful Life prediction using the NASA C-MAPSS dataset.

The proposed framework included preprocessing, sequence generation, recurrent neural network modeling, ensemble learning, and Group K-Fold Cross Validation.

The following chapter presents the experimental results, model comparisons, and detailed performance analysis across all dataset subsets.

6 Results and Discussion

6.1 Introduction

This chapter presents the experimental results obtained from the implemented machine learning and deep learning models for Remaining Useful Life (RUL) prediction using the NASA C-MAPSS dataset. The proposed models were evaluated on the four benchmark subsets (FD001, FD002, FD003, and FD004) to assess prediction accuracy, robustness, and generalization capability under different operational conditions and fault modes.

The implemented models include:

- BiLSTM
- GRU with Attention
- XGBoost
- Ensemble Learning

The evaluation process utilized Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE), where lower values indicate better prediction performance.

Two complementary evaluation protocols are reported in this chapter. The **held-out test evaluation** (Tables 5–8) uses a single engine-level 80/20 split of the training data into training and validation partitions, trains each model for 20 epochs with model checkpointing on the validation set, and reports performance on the official C-MAPSS test files using the provided ground-truth RUL values. The **Group K-Fold cross-validation** (Table 9) partitions the training-set engines into $k = 5$ disjoint folds and trains a fresh model on each fold (10 epochs per fold), reporting the mean and standard deviation of the per-fold validation metrics. The two protocols measure different properties: the first measures generalization to the official test set, while the second measures the stability of the training procedure across different engine subsets.

6.2 FD001 Results

FD001 represents the simplest subset because it contains a single operating condition and a single fault mode. Therefore, the models achieved their strongest overall performance on this dataset.

The final test results for FD001 are presented in Table 5.

Table 5FD001 Test Results

Model	RMSE	MAE
BiLSTM	11.52	7.58
GRU + Attention	12.06	8.18
XGBoost	11.15	8.19
Ensemble	10.60	7.38

The Ensemble model achieved the best overall performance with RMSE approximately 10.60 and MAE approximately 7.38. This improvement is attributed to combining temporal sequence learning from recurrent neural networks with feature-based learning from XGBoost.

BiLSTM also demonstrated strong performance due to its ability to process degradation sequences bidirectionally and capture long-term temporal dependencies.

6.3 FD002 Results

FD002 contains multiple operating conditions with a single fault mode, making it considerably more difficult than FD001. The increased operational variability introduces additional complexity in modeling engine degradation patterns.

The final test results obtained on FD002 are presented in Table 6.

Table 6FD002 Test Results

Model	RMSE	MAE
BiLSTM	19.45	13.91
GRU + Attention	21.03	15.64
XGBoost	20.14	15.82
Ensemble	18.62	14.36

The Ensemble model achieved the best overall performance on FD002 with RMSE approximately 18.62. The results indicate that combining predictions from multiple architectures improves robustness under varying operational conditions.

Compared to FD001, all models experienced a noticeable increase in prediction error due to the higher complexity of the dataset. This demonstrates the significant impact of operational variability on Remaining Useful Life prediction accuracy.

6.4 FD003 Results

FD003 contains a single operating condition but multiple fault modes. Although the dataset contains more fault diversity than FD001, the absence of multiple operational settings allowed the models to maintain relatively strong predictive performance.

The obtained test results are shown in Table 7.

Table 7FD003 Test Results

Model	RMSE	MAE
BiLSTM	13.06	7.40
GRU + Attention	11.85	7.18
XGBoost	15.64	10.53
Ensemble	12.24	7.55

The GRU with Attention model achieved the strongest overall performance on FD003 with RMSE approximately 11.85 and MAE approximately 7.18.

The attention mechanism improved the model’s ability to focus on important degradation features within the sequential sensor data, resulting in improved prediction accuracy.

The results also show that deep learning models outperform XGBoost on datasets with strong temporal degradation behavior.

6.5 FD004 Results

FD004 is considered the most challenging subset because it contains both multiple operating conditions and multiple fault modes simultaneously.

The final test results are presented in Table 8.

Table 8FD004 Test Results

Model	RMSE	MAE
BiLSTM	20.52	11.39
GRU + Attention	19.18	10.54
XGBoost	17.53	11.89
Ensemble	17.44	10.51

The Ensemble model achieved the best overall performance on FD004 with RMSE approximately 17.44 and MAE approximately 10.51.

Despite the increased dataset complexity, the ensemble approach maintained stable prediction capability by combining temporal sequence learning and feature-based learning.

Interestingly, XGBoost achieved competitive performance on FD004, suggesting that tree-based learning methods can generalize effectively under highly complex operating environments.

6.6 Group K-Fold Cross Validation Results

To further evaluate model robustness and generalization capability, Group K-Fold Cross-Validation with $k = 5$ was implemented across all four C-MAPSS subsets. Unlike conventional K-Fold validation, Group K-Fold ensures that engine units appearing in the training set do not also appear in the validation set within any fold, preventing leakage between engine trajectories.

To control the total computational cost of training $5 \times 4 = 20$ separate model configurations, each fold was trained for 10 epochs rather than the 20 epochs used for the held-out test experiments. All other training settings (architectures, optimizer, learning rate, batch size, loss function, gradient clipping, and model checkpointing) remained identical to the main experiments.

The average cross-validation results are presented in Table 9.

Table 9 Group K-Fold Cross Validation Results

Subset	Model	MAE _{mean}	RMSE _{mean}	RMSE _{std}
FD001	GRU + Attention	9.64	13.55	1.06
FD001	Ensemble	10.55	14.00	0.64
FD001	BiLSTM	9.96	14.28	0.71
FD001	XGBoost	14.42	19.33	0.60
FD002	Ensemble	14.15	18.82	1.90
FD002	GRU + Attention	14.39	19.50	1.53
FD002	XGBoost	15.49	20.32	1.44
FD002	BiLSTM	14.94	20.81	1.69
FD003	BiLSTM	8.15	12.82	2.13
FD003	Ensemble	8.90	13.06	2.00
FD003	GRU + Attention	8.65	13.08	1.72
FD003	XGBoost	12.35	17.68	1.93
FD004	XGBoost	12.98	18.40	1.28
FD004	Ensemble	12.80	18.60	1.51
FD004	GRU + Attention	13.92	20.77	1.53
FD004	BiLSTM	13.70	20.82	1.51

The cross-validation results demonstrate relatively stable performance across different folds, as indicated by the moderate RMSE standard deviation values.

For FD001, the GRU with Attention model achieved the best average RMSE during cross-validation, indicating strong generalization capability under simpler operational settings.

For FD002, the Ensemble model achieved the best average RMSE, demonstrating the advantage of combining multiple prediction strategies under varying operating conditions.

For FD003, the BiLSTM model achieved the strongest cross-validation performance due to its ability to capture long-term temporal dependencies within sequential degradation patterns.

For FD004, XGBoost achieved the best average RMSE during cross-validation, suggesting that feature-based machine learning approaches may generalize effectively under highly complex operational environments.

Overall, the cross-validation experiments confirm that model performance varies significantly depending on dataset complexity, operational variability, and fault diversity.

6.7 Comparison with Previous Studies

The obtained results were compared with several previous studies discussed in Chapter 2.

Table ?? presents a comparison between the proposed approach and selected related work.

Table 10 Comparison with Previous Studies on FD001

Study	Model	Dataset	RMSE
Deng & Zhou [8]	CNN-LSTM-Attention	FD001	15.977
Gupta et al. [10]	RFR / SVR / LR	FD001	19.007
Rath et al. [20]	BiLSTM w/ Change-Point	FD001	18.71
This work	Ensemble (BiLSTM + GRU+Attn + XGBoost)	FD001	10.60

The proposed Ensemble model achieved lower RMSE values compared to several previous studies. However, direct comparison should be interpreted carefully because different studies may utilize different:

- preprocessing strategies
- train-test splitting methods
- feature engineering techniques
- RUL labeling methods
- evaluation settings

Several methodological choices likely contribute to this gap. First, training uses Huber loss ($\delta = 10$) rather than MSE, which moderates the gradient contribution of the clipped healthy-phase samples and stabilizes optimization. Second, the StandardScaler is fitted exclusively on the training engines of each split, with no test-set statistics leaking into the scaling parameters. Third, model checkpointing — selecting the epoch with the lowest

validation MAE rather than the final epoch — yields a more conservative final model. Fourth, the ensemble combines three architecturally distinct learners: BiLSTM captures bidirectional temporal context, GRU+Attention performs selective temporal focus, and XGBoost models non-linear feature interactions on the most recent sensor state. Equal-weight averaging across these models reduces prediction variance without privileging any single inductive bias.

It must be acknowledged that direct comparisons across published studies are inherently limited by differences in train/test splitting, sensor selection, sequence-window length, RUL clipping thresholds, and hyperparameter configurations. The magnitude of the reported improvement should therefore be interpreted with appropriate caution.

6.8 Discussion

The experimental results demonstrate that recurrent neural networks remain highly effective for Remaining Useful Life prediction due to their ability to capture temporal degradation behavior from sequential sensor measurements.

BiLSTM demonstrated strong performance by learning bidirectional temporal dependencies, while the GRU with Attention model achieved stable cross-validation performance through selective temporal focus.

XGBoost demonstrated that machine learning approaches remain highly competitive, especially under complex operational conditions such as FD004.

The Ensemble model consistently achieved strong overall performance because it combines the strengths of both deep learning and machine learning approaches.

The experiments also confirmed that increasing operational complexity significantly impacts Remaining Useful Life prediction accuracy. FD002 and FD004 produced considerably higher prediction errors due to multiple operating conditions and fault modes.

6.9 Conclusion

This chapter presented the experimental results and performance analysis of the implemented models across the four NASA C-MAPSS subsets.

The proposed Ensemble model achieved the strongest overall performance on FD001, while Group K-Fold Cross Validation demonstrated stable model behavior and strong generalization capability.

The experiments also confirmed that dataset complexity significantly affects Remaining Useful Life prediction performance, particularly under multiple operating conditions and multiple fault modes.

The following chapter presents the final conclusions of the project and discusses future research directions and possible improvements.

7 Conclusion and Future Work

7.1 Introduction

This chapter presents the final conclusions of the project along with the main challenges encountered during development and several possible directions for future improvements.

The project focused on developing a predictive maintenance framework for Remaining Useful Life (RUL) estimation using the NASA C-MAPSS turbofan engine dataset. Multiple machine learning and deep learning models were implemented and evaluated under different operational conditions and fault modes.

The experiments demonstrated the importance of temporal sequence modeling, feature engineering, ensemble learning, and proper validation techniques for improving predictive maintenance performance.

7.2 Conclusion

This project successfully developed and evaluated multiple approaches for Remaining Useful Life prediction using multivariate aircraft engine sensor data.

Several preprocessing and feature engineering techniques were applied, including normalization, sensor selection, sequence generation, and piecewise linear RUL labeling. These preprocessing stages improved data quality and enhanced model training stability.

Different machine learning and deep learning architectures were implemented, including BiLSTM, GRU with Attention, XGBoost, and Ensemble Learning. The models were evaluated across all four NASA C-MAPSS subsets (FD001, FD002, FD003, and FD004), which contain varying operational conditions and fault complexities.

The experimental results demonstrated that recurrent neural networks are highly effective for learning temporal degradation behavior from sequential sensor measurements. BiLSTM and GRU with Attention achieved strong performance due to their ability to capture long-term dependencies within engine degradation sequences.

The Ensemble model achieved the strongest overall performance on multiple datasets by combining the advantages of deep learning and machine learning approaches. The ensemble strategy reduced prediction variance and improved robustness under complex operational conditions.

The results also showed that dataset complexity significantly affects prediction accuracy. Subsets containing multiple operational conditions and fault modes produced higher prediction errors due to increased degradation variability.

Additionally, Group K-Fold Cross Validation demonstrated relatively stable model performance across validation folds, confirming the robustness and generalization capability of the implemented framework.

Overall, the project demonstrates that combining temporal sequence learning, feature-based learning, and ensemble strategies can significantly improve Remaining Useful Life prediction performance for predictive maintenance applications.

Based on the experimental evaluation, the Ensemble model is recommended for deployment in scenarios that prioritize accuracy across varying operational conditions. For deployment under tighter computational constraints, the GRU with Attention model offers the best single-model trade-off between predictive accuracy and parameter count.

7.3 Challenges and Limitations

Several challenges and limitations were encountered during the development of this project.

One of the primary challenges was the complexity of the NASA C-MAPSS dataset. The dataset contains multivariate time-series sensor measurements collected under varying operational conditions and fault modes, making degradation modeling significantly more difficult.

Another challenge involved selecting appropriate preprocessing strategies and hyperparameters for the implemented deep learning architectures. Recurrent neural networks require careful tuning of sequence length, hidden layers, dropout rates, batch size, and learning rates to achieve stable convergence and avoid overfitting.

The project also faced computational limitations during training and cross-validation experiments. Deep learning architectures require considerable computational resources and training time, particularly when evaluated across multiple dataset subsets and validation folds.

Although the implemented models achieved strong performance, several limitations still exist. The experiments were conducted using simulated benchmark datasets rather than real-world industrial data. Real operational environments may contain additional complexities such as sensor noise, missing values, unexpected failures, and changing environmental conditions.

Furthermore, the ensemble method used in this project relies on *simple (equal-weight) averaging* and may not fully exploit more advanced ensemble strategies. Additional improvements may still be possible through stacking, learned weighting, or Bayesian model averaging. A further limitation is that the originally proposed LLM-driven conversational maintenance assistant, listed as an objective in Section 1.4.2, was not implemented within the scope of this graduation project. Available time and computational resources were prioritized for evaluating all four C-MAPSS subsets and performing Group K-Fold cross-validation across them. The integration of a conversational assistant is discussed as a priority direction in Section 7.4.

7.4 Future Work

Several future improvements can further enhance the proposed predictive maintenance framework.

The most immediate extension is the development of the originally planned LLM-driven conversational maintenance assistant. This component would wrap the trained ensemble model behind a natural-language interface, allowing maintenance engineers to query engine health status, request natural-language explanations of individual RUL predictions, and receive maintenance scheduling recommendations through plain-language interaction.

A second potential improvement involves implementing Transformer-based architectures for Remaining Useful Life prediction. Transformer models have recently demonstrated strong performance in sequential learning tasks and may improve long-term temporal dependency modeling. Another possible direction is integrating Explainable Artificial Intelligence (XAI) techniques such as SHAP and LIME. These methods would improve model interpretability and help maintenance engineers understand how specific sensor measurements influence RUL predictions.

Future work may also explore transfer learning and domain adaptation techniques to improve model generalization across different engine types and operational environments with limited training data.

In addition, real-time deployment can be investigated by integrating the predictive maintenance framework with live sensor streams and monitoring dashboards. This would allow maintenance engineers to continuously monitor engine health and schedule maintenance operations proactively.

Future studies may also investigate probabilistic RUL estimation and uncertainty quantification methods to provide confidence intervals for predictions rather than single deterministic outputs.

Finally, additional research can focus on hybrid ensemble optimization methods and reinforcement learning approaches for intelligent maintenance scheduling and decision-making systems.

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